

PAUL MITCHELL

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AI, DATA ANALYTICS & SYSTEMS ENGINEERING ANALYST

PROFESSIONAL SUMMARY

William & Mary Computer Science M.S. candidate (3.95 GPA, AI/ML concentration) and B.S. graduate specializing in reproducible data engineering, applied machine learning pipelines, and low-latency AI infrastructure. Combines hands-on Linux systems administration (100+ instructional nodes) with production NLP fine-tuning (DistilBERT, BERTopic), predictive modeling (LightGBM), and pre-inference LLM middleware. Experienced in translating complex statistical corpora into commercial decision tools and executive dashboards.

EDUCATION

College of William & Mary — M.S. in Computer Science (AI/ML & Systems) Expected Dec 2026 · GPA: 3.95
College of William & Mary — B.S. in Computer Science Graduated Dec 2024
Relevant Coursework: Machine Learning Systems, Advanced Software Engineering, Deep Learning, Distributed Systems, Data Mining

TECHNICAL SKILLS & INFRASTRUCTURE

Data Science & ML: PyTorch, scikit-learn, LightGBM, DistilBERT, Sentence-BERT, BERTopic, Hugging Face, statistical modeling, pandas, NumPy
Data Engineering & Cloud: Python (ETL pipelines, spaCy), SQL (PostgreSQL), AWS (S3, EC2, Bedrock), DVC, Apache Kafka, Redis, Docker, Git CI
Analytics & Dashboards: Streamlit, Grafana, Prometheus, OpenTelemetry, Excel (Advanced Modeling, Pivot Tables), business telemetry
Software & Systems: FastAPI, RESTful microservices, Linux (Ubuntu/RHEL), Bash, C/C++, Java, test-driven development, system hardening

TECHNICAL & PROFESSIONAL EXPERIENCE

Data Analytics & AI Engineering Extern | Extern (Amazon-Sponsored Project) | Remote Jun 2025 – Sep 2025
• Engineered automated ETL pipelines with Python, spaCy, and AWS S3 to clean, normalize, and structure ~20,000 employee review records, versioning datasets via DVC.
• Fine-tuned a DistilBERT text classification model (F1 = 0.89) and applied BERTopic clustering to quantify five root causes of operational turnover across unstructured text.
• Programmed a LightGBM predictive attrition engine and deployed an interactive Streamlit analytics platform delivering real-time decision metrics for program leadership.
• Prototyped a retrieval-augmented insight agent querying structured survey corpora to provide on-demand retention recommendations.

IT Administrator | William & Mary Technology Support Center | Williamsburg, VA Feb 2025 – Present
• Architected telemetry pipelines using Prometheus and Grafana, capturing real-time hardware metrics across departmental compute clusters to identify resource bottlenecks.
• Administer 100+ Linux compute nodes; automate system provisioning, user isolation, and software stack rollouts via Puppet and Bash scripting.
• Maintained relational inventory tracking mechanisms in Excel, reconciling software licenses and hardware lifecycles across 100+ lab workstations.
• Conducted capacity utilization evaluations across multi-user nodes, authoring benchmark summaries to inform future hardware allocation.
• Configured network interfaces, static IP routing, and firewall policies ensuring uninterrupted instructional access across all lab pods.

KEY TECHNICAL & SYSTEMS PROJECTS

ShadowPrompt: High-Throughput Pre-Inference AI Security Proxy | Python, FastAPI 2026 · pjmittchell7.github.io/shadowprompt
• Engineered an in-memory streaming proxy evaluating raw token streams in 0.038ms P99 latency, neutralizing zero-width steganography and prompt injection prior to LLM tokenization.
• Integrated deceptive honeypot sandboxes seeded with canary credentials to analyze adversary behavior; outputted structured STIX 2.1 telemetry feeds for enterprise SIEMs.

ChaosLedger: Distributed Financial Stream Engine under Simulated AWS Partitions | Python, Kafka, Redis 2026
• Architected a financial stream engine simulating 100k tx/min under chaos-injected 200ms latency spikes; guaranteed \$0.00 balance discrepancy using atomic Redis idempotency.
• Engineered adaptive consumer-lag backpressure preventing Kafka partition rebalance storms, maintaining sub-6ms P99 processing latency.

Automotive CAN Bus Intrusion Detection Pipeline | Python, scikit-learn 2025
• Processed and analyzed high-frequency automotive CAN packet streams, transforming raw binary message frames into structured features for attack classification.
• Achieved 99.4% detection accuracy for simulated denial-of-service and message injection attacks using lightweight random forest classifiers.